drawing the line

digitalisierung und programmierung · input, session 4 agenda

1. a world without steps
2. drawing your own lines

a world without steps

is this digital?

image: ai-generated (gpt-image-2)

the part no program controls

the middle zone

light is analog: between any two brightnesses lies another.

the middle zone

light is analog: between any two brightnesses lies another.

your program needs states: finitely many, clearly apart.

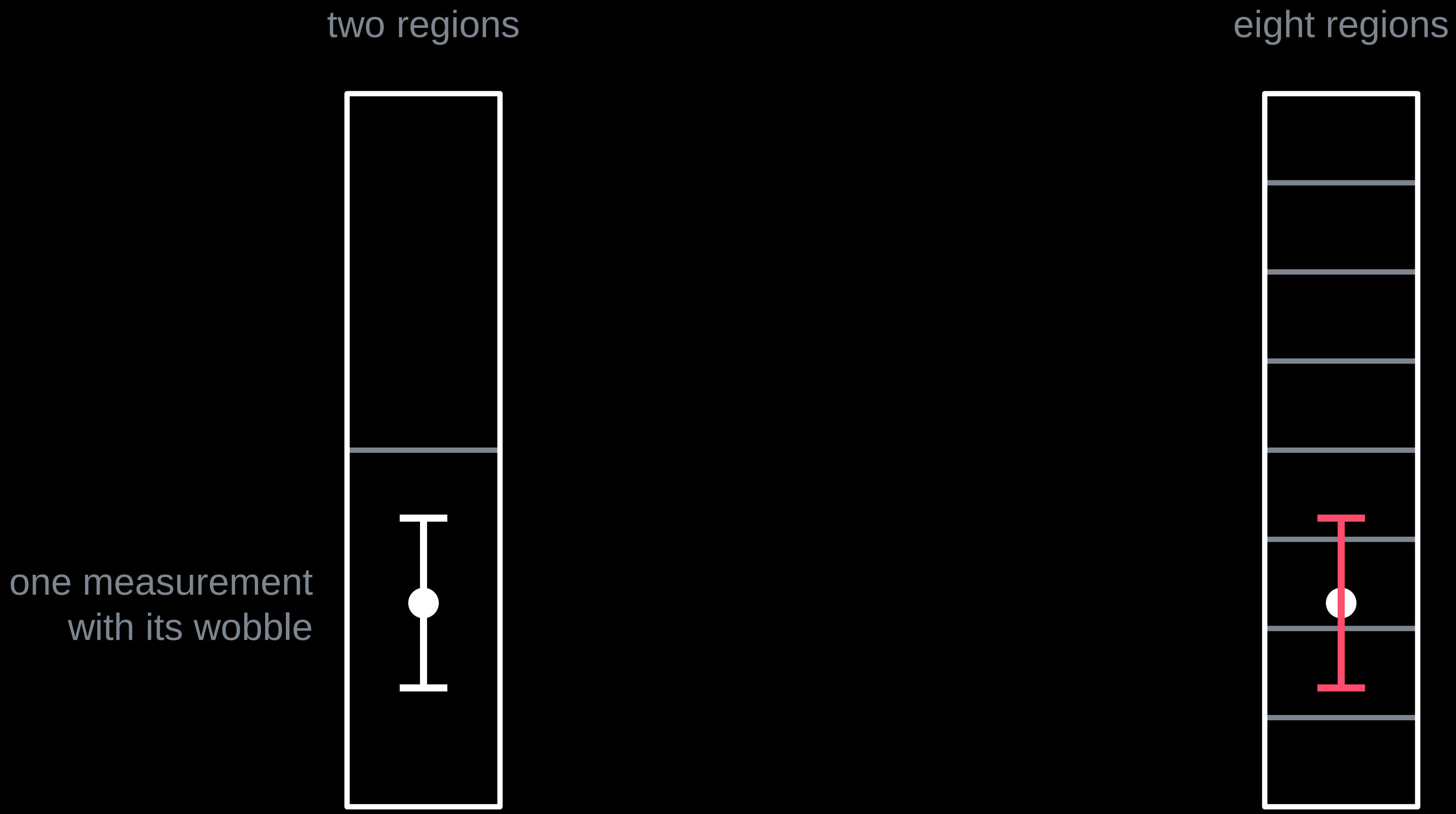
digital is not the same as electronic

digital	analog
split-flap board	slide rule
light switch	dial thermometer
abacus	dimmer
a die	hourglass

not one of them contains a chip.

**digital means finitely many states,
and nothing in between counts.**

same noise, more regions



the noise did not grow. the margin shrank.

digitising throws information away. on purpose.

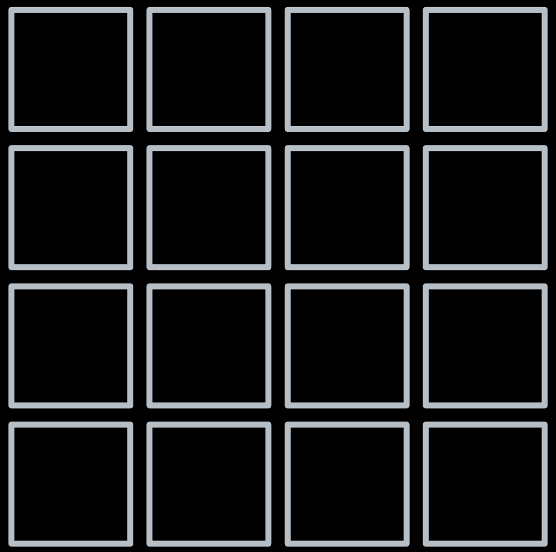
which exact value it was inside the region: gone for good.

choose what to throw away:
`winf-hsos.github.io/lifi-concept-demos`

digitising takes two cuts

cut 1: sampling

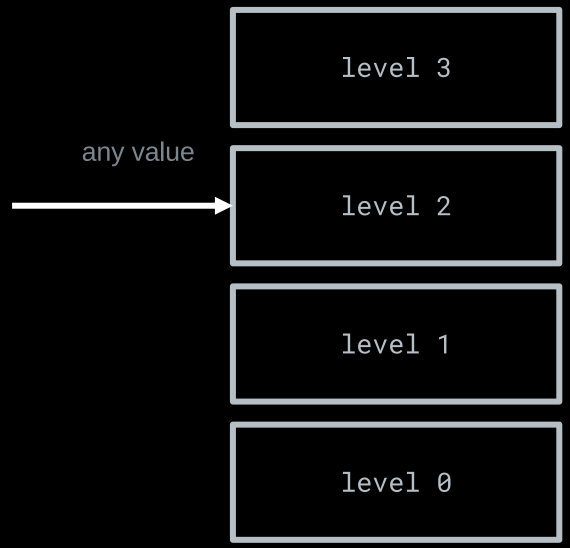
slice the area into points



how many points? that is the resolution

cut 2: quantization

slice the value into levels



how many levels? that is the colour depth

sampling slices the axis, quantization slices the value. both throw detail away — on purpose.

points × bits per point = file size

$$64 \times 64 \text{ pixels} \times 24 \text{ bit} = 98{,}304 \text{ bit} = 12 \text{ KB}$$

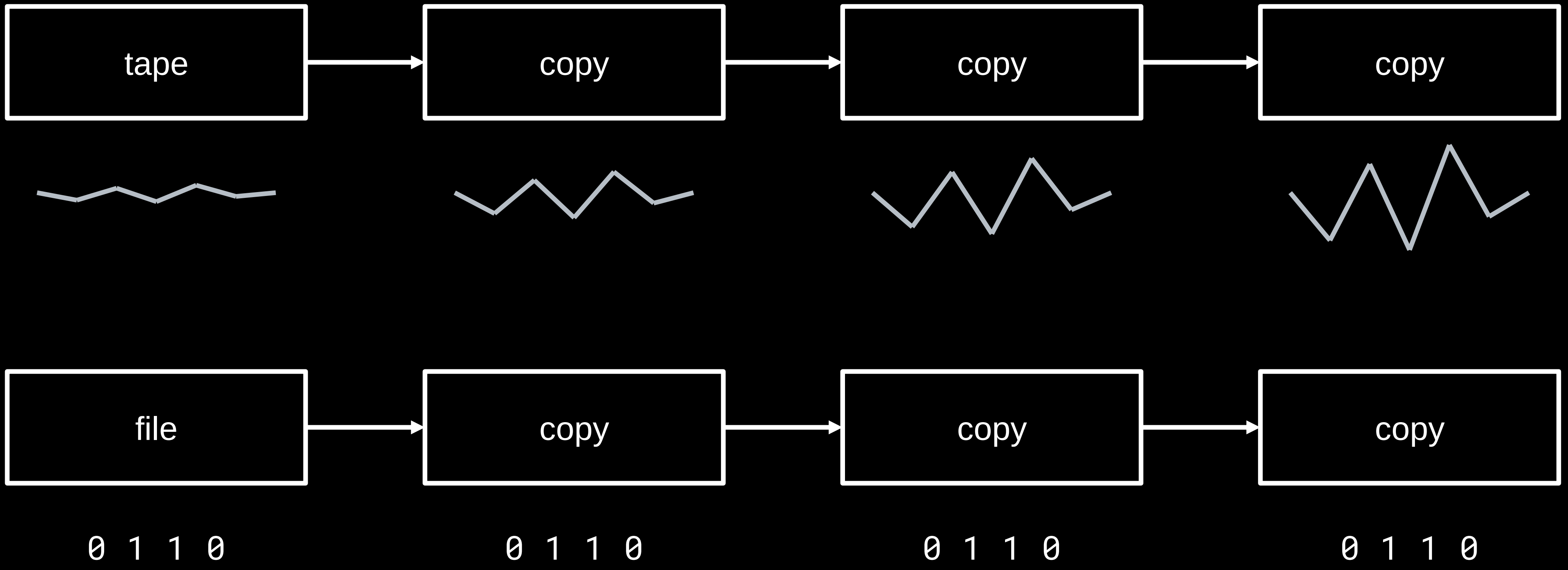
at 30 bit/s over the light link: about 55 minutes

$$64 \times 64 \text{ pixels} \times 1 \text{ bit} = 4{,}096 \text{ bit} = 512 \text{ bytes}$$

at 30 bit/s: about 2 minutes – same picture, in black and white

resolution and depth are levers: what you keep, you pay for in bits.

what the thrown-away precision buys



copy it yourself:

winf-hsos.github.io/lifi-concept-demos

every digital copy is a rebirth from clean states.

drawing your own lines

```
value = sensor.get_clear()
if value > 250:
    symbol = "bright"
else:
    symbol = "dark"
```

the sensor delivers hundreds of possible numbers.
this line turns them into two symbols.

how many colours are safe? your measurements decide.

two are safe. eight are fast. somewhere in between sits your alphabet, and only your own data knows where.